

$$J_{CE} = - \sum_{k=1}^m y_k \log \hat{y}_k$$

$$\begin{aligned} J(\hat{y}, y) &= \frac{1}{m} \sum_{i=1}^m J_{CE}^{(i)} \\ &= \frac{1}{m} \sum_{i=1}^m \sum_{k=1}^K y_k^{(i)} \log \hat{y}_k^{(i)} \end{aligned}$$

$$\frac{\partial J}{\partial \hat{y}_k} = - \frac{y_k}{\hat{y}_k}$$

SOFTMAX

$$\sigma(z)_k = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}} \rightarrow \hat{y}_k = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}$$

$$\begin{aligned} \frac{\partial \hat{y}_k}{\partial z_j^{(i)}} &= \frac{0 \cdot \sum e^{z_j} - e^{z_k} e^{z_j}}{(\sum e^{z_j})^2} \quad j \neq k \\ &= \frac{-e^{z_k} e^{z_j}}{(\sum e^{z_j})^2} \\ &= -\frac{e^{z_k}}{\sum e^{z_j}} \left(\frac{e^{z_j}}{\sum e^{z_j}} \right) \\ &= -\hat{y}_k \hat{y}_j \end{aligned}$$

$$\begin{aligned} \frac{\partial \hat{y}_k}{\partial z_k^{(i)}} &= \frac{e^{z_k} \sum e^{z_j} - e^{z_k} e^{z_k}}{(\sum e^{z_j})^2} \\ &= \frac{e^{z_k} \sum e^{z_j}}{(\sum e^{z_j})^2} - \frac{e^{z_k} e^{z_k}}{(\sum e^{z_j})^2} \\ &= \frac{e^{z_k}}{\sum e^{z_j}} - \left(\frac{e^{z_k}}{\sum e^{z_j}} \right)^2 \\ &= \hat{y}_k - \hat{y}_k^2 \\ &= \hat{y}_k (1 - \hat{y}_k) \end{aligned}$$

$$\frac{\partial \hat{y}_k}{\partial z_j^{(i)}} = \begin{cases} \hat{y}_k (1 - \hat{y}_k) & \text{if } j=k \\ -\hat{y}_k \hat{y}_j & \text{if } j \neq k \end{cases}$$

$$= \hat{y}_k (\delta_{jk} - \hat{y}_j) \quad \delta = 1 \text{ if } j=k, \text{ otherwise } 0$$

$$\begin{aligned}
\frac{\partial \mathcal{J}}{\partial z_j^{(2)}} &= \sum_k \frac{\partial \mathcal{J}}{\partial \hat{y}_k} \frac{\partial \hat{y}_k}{\partial z_j^{(2)}} \\
&= \sum_k \frac{-y_k}{\hat{y}_k} \hat{y}_k (\delta_{jk} - \hat{y}_j) \\
&= \sum_k -y_k (\delta_{jk} - \hat{y}_j) \\
&= -\sum_k y_k \delta_{jk} + \sum_k y_k \hat{y}_j \\
&= -y_j + \underbrace{\sum_k y_k}_{=1 \text{ since } y_k \text{ is one-hot}} \hat{y}_j \\
&= \hat{y}_j - y_j
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \mathcal{J}(\hat{y}, y)}{\partial z^{(2)}(i)} &= \frac{\partial \mathcal{J}}{\partial z^{(2)}(i)} \\
&= \frac{1}{m} \left[\frac{\partial \mathcal{J}^{(1)}}{\partial z^{(2)}(i)} + \dots + \frac{\partial \mathcal{J}^{(i)}}{\partial z^{(2)}(i)} + \dots + \frac{\partial \mathcal{J}^{(m)}}{\partial z^{(2)}(i)} \right] \\
&= \frac{1}{m} \left[0 + \dots + \frac{\partial \mathcal{J}^{(i)}}{\partial z^{(2)}(i)} + \dots + 0 \right] \\
&= \frac{1}{m} (\hat{y}_i - y_i)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \mathcal{J}(\hat{y}, y)}{\partial z^{(2)}} &= \begin{bmatrix} \frac{\partial \mathcal{J}}{\partial z^{(2)}(1)} \\ \vdots \\ \frac{\partial \mathcal{J}}{\partial z^{(2)}(m)} \end{bmatrix} \\
&= \frac{1}{m} [\hat{y} - y]
\end{aligned}$$